

Results

01 · KRUSKAL-WALLIS

nonparametric 3+ group comparison

Table 1. Rank summary

Group	N	Median	IQR	Mean rank
control	20	33.55	9.63	24.90
drug_a	20	36.50	11.97	28.15
drug_b	20	38.70	6.60	38.45

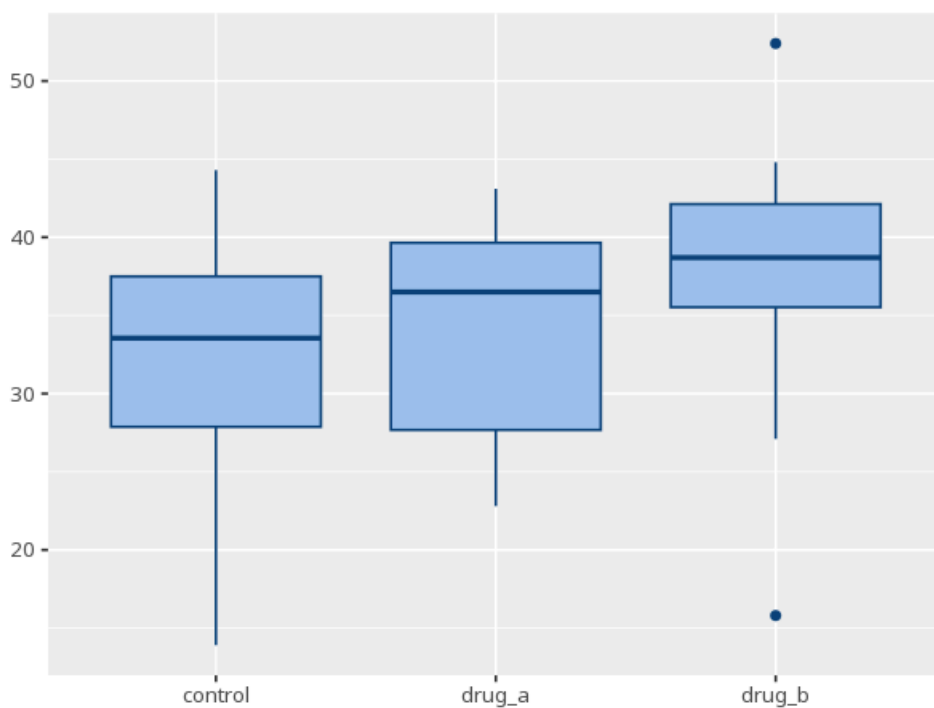
Table 2. Kruskal-Wallis test

H	df	p	ε² [95% CI]
6.56	2	.038	0.11 [0.03, 1.00]

Table 3. Dunn post-hoc

Pair	Z	P _{adj}
control - drug_a	0.59	.556
control - drug_b	2.45	.042
drug_a - drug_b	1.87	.124

Figure. Distribution by group



Understanding the numbers

df. The degrees of freedom for the Kruskal-Wallis test, based on the number of groups. Here, $df = 2$.

ϵ^2 . The effect size for the Kruskal-Wallis test – the proportion of rank variance explained by group membership. Here, $\epsilon^2 = 0.11$ [0.03, 1.00].

H. The Kruskal-Wallis test statistic, testing whether any group's ranks differ systematically from the others. Here, $H(2) = 6.56$.

IQR. The middle 50% spread of scores within each group (descriptive, not part of the test itself). Here, the IQR ranges from 6.60 in drug_b to 11.97 in drug_a.

Mean rank. The average rank of each group's scores when all scores are ranked together – the quantity H is calculated from. Here, the mean rank ranges from 24.90 in control to 38.45 in drug_b.

Median. The middle value of each group's scores. Here, the median ranges from 33.55 in control to 38.70 in drug_b.

N. The total number of observations used across all groups. Here, $N = 60$.

p. The probability of rank differences this large (or larger) across groups if all groups truly had the same distribution. Here, $p = .038$ - below your significance threshold, at least one group differs.

p adj. The Dunn post-hoc adjusted p-value for a specific pair of groups. Here, the strongest pairwise difference is control - drug_b, $p \text{ adj} .042$.

Z. The Dunn post-hoc test statistic comparing a specific pair of groups' mean ranks. Here, for control - drug_b, $Z = 2.45$.

How to read this test

The nonparametric counterpart to one-way ANOVA. The H/p tests whether any group tends to have systematically higher or lower values (stochastic dominance) — read it as a "median difference" only when the groups have similar distribution shapes. If significant, Dunn's post-hoc shows which pairs differ, with adjusted p-values. ϵ^2 is the effect size. Your significance threshold (α) is 0.05.

APA template: A Kruskal-Wallis test gave $H(2)=6.56$, $p = .038$, $\epsilon^2=.11$ [.03, 1.00].

Statistical basis: Kruskal-Wallis H test (nonparametric counterpart to one-way ANOVA) (Kruskal, W. H., Wallis, W. A. (1952). "Use of ranks in one-criterion variance analysis." Journal of the American Statistical Association, 47(260), 583-621.); Dunn's post-hoc pairwise comparisons (Kassambara A (2023). rstatix: Pipe-Friendly Framework for Basic Statistical Tests. R package.); ϵ^2 effect size (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." Journal of Open Source Software, 5(56), 2815.). Full references in the exported CITATIONS.txt.

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